

XUAN B. TRAN

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Education

Brighton College School - IB Diploma Programme

Hanoi

Predicted IB Score: 28/45

Sep 2024 -

Higher Level: Economics (5), Physics (5), Vietnamese Literature (5)

June 2027

Standard Level: English A Language & Literature (4), Mathematics AA (4), Chemistry (5)

STANDARDIZED TESTS

IELTS: 8.5

Oct 2025

SAT: 1400

Sep 2025

Honors & Awards

Bronze Award – British Physics Olympiad (BPhO) – Round 1

Awarded by the BPO Committee for outstanding performance in the regional qualifying round.

Gold Medal – Artificial Intelligence Global College (AIGC)

Developed a machine learning model to identify diabetic symptoms by analyzing retinography scans.

Gold Medal – 1-Idea 1-World International Innovation Design Competition (1I1W)

Developed an IoT system to monitor and manage urban greenery for smart city infrastructure.

Research Experience

Applications and Impacts of Internet of Things (IoT) in Logistics Industry in Vietnam - [\[Detail\]](#)

Jul 2025

Published in the IJSRM, Vol. 13, Issue 07, 2025

Key Contributions:

- Co-authored a study on Vietnam's logistics digital transformation, highlighting a 68% IoT adoption rate among large-scale firms.
- Evaluated IoT performance in cold chain management, specifically identifying a 15% reduction in product damage through real-time monitoring.
- Conducted numerical analysis to forecast Vietnam's IoT sector valuation at \$ 3.13 billion by 2025, with a 12.7% annual growth rate.

Classification of Fruits using Image Processing and Deep Neural Networks - [\[Detail\]](#)

Dec 2024

Published in the IJARCS, Vol. 15, No. 6, Nov–Dec 2024

Key Contributions:

- Applied advanced deep neural networks (AlexNet, ResNet-50, MobileNetV3) to fruit image classification, achieving up to 94% accuracy.
- Enhanced dataset quality and balance through data augmentation techniques (rotation, color adjustment, Gaussian noise).
- Optimized classification speed with MobileNetV3, enabling deployment on resource-constrained devices for real-world applications in agriculture and industry.

Personal Project

Product name: AI-Powered Air Quality Monitoring & Predictive Analysis System

2025

Key Features / Highlights:

- Designed and assembled a sensor node using Arduino/ESP32 to monitor real-time environmental parameters, including $PM_{2.5}$, PM_{10} , CO_2 , temperature, and humidity.
- Implemented a Machine Learning model (linear regression or LSTM) to analyze historical trends and predict potential air-pollution spikes in specific urban areas.
- Developed a centralized dashboard to visualize data patterns, enabling users to track air quality indices (AQI) via a web or mobile interface.

Leadership Experience

Academic Board Leader

- Designed 15+ academic modules and 4 mock exams for 200+ students.
- Directed a 12-member team to run bi-weekly technical workshops with 90% attendance.
- Coordinated with faculty to align projects with STEM curriculum, boosting competition enrollment by 30%.

Lead Organizer – “Hope After the Storm” Charity Campaign

- Led 3 relief expeditions to Central Vietnam, delivering 500+ school kits and winter clothing.
- Raised \$2,000 in 3 weeks through a targeted crowdfunding campaign (125% of goal).
- Managed volunteer teams to safely distribute 2 tons of aid to 150+ households.

Social Work & Activities

Volunteer - "Tet Warmth for Children 2025"

2025

- Orchestrated 3 major relief expeditions to mountainous regions, providing 500+ "Back-to-School" kits to underprivileged students.
- Managed the distribution of 2 tons of relief goods across challenging terrains to support 150+ households in remote areas.

Competitive Debater - [Tên giải Debate]

202x

- Developed 20+ comprehensive case files on complex socio-economic topics for a national-level tournament with 40+ teams.
- Delivered persuasive speeches under strict 7-minute time constraints, achieving an 85% success rate in rebuttal rounds.
- Conducted 50+ hours of in-depth research to construct logical frameworks, leading the team to the Quarter-finals.

Volunteer - SOS Children’s Village Hanoi

2025

- Participated in community service activities supporting orphaned and disadvantaged children.
- Assisted with educational and recreational programs to foster a supportive environment.

Volunteer - Xanh Pon Hospital

2026

- Contributed to volunteer work aiding pediatric patients and hospital staff.
- Helped organize donation drives and provided direct support to families in need.

Volunteer - Green Horizon Project, Tuyên Quang

2026

- Joined environmental volunteer initiatives focused on reforestation and sustainable living.
- Supported awareness campaigns promoting eco-friendly practices in rural communities.

Core Skills

Engineering & IoT Development: Proficient in designing embedded systems using Arduino/ESP32, integrating sensors (PM2.5, CO2), and implementing NB-IoT networking for real-time monitoring.

Data Analysis & AI: Experienced in leveraging Machine Learning models (Linear Regression/LSTM) and Big Data analytics to identify operational trends and predict environmental patterns.

Technical Design & Documentation: Skilled in using CAD for 3D printing prototypes, alongside professional graphic tools like Canva and Photoshop for technical reporting and marketing.

Strategic Project Leadership: Proven ability to manage \$2,000+ budgets, coordinate multi-channel fundraising, and lead multidisciplinary teams to meet strict project deadlines.